



Department of Computer Science and Engineering

Academic Year 2022– 2023 (Even Semester)

Degree, Semester & Branch: VI Semester – B.E. CSE

Course Code & Title: CS8601 Mobile Computing

Name of the Faculty member (s): Dr.R.Venkatesh, ASP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit II / GPRS - Architecture

- **Course Outcome:** CO2

- **Topic Learning Outcome:** TLO7

- **Activity Chosen:** One Minute Paper

- **Justification:**

The topic “GPRS Architecture” contains theory part and its architecture. This activity provides students with a strong understanding of this concept. A minute paper is a brief writing task in which students respond to questions posed by the teacher about the topics covered in class. These questions encourage students to reflect on the lessons they've learned and provide feedback on their understanding. This assists teachers in preparing for the following class. Its main advantage is that it provides immediate feedback on whether the instructor's and students' main ideas are the same.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

- The instructor explained the specific concepts/topic in the classroom in 35 minutes.
- At the end of the class, students were instructed to write about the topic addressed in class based on the discussion.
- The first 5 minutes the students revised the concepts and last 10 minutes in class, students expressed and individually wrote what they had learned. The figure 1 shows the students actively writing the GPRS and its architecture and what they had learned
- This activity shows whether students understand the concepts, materials and are actively participating in the lesson.
- The figure 2 shows the sample One Minute Paper of Saravanajothi G and Venesha L.



- CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO9	PO10	PSO1
CO2	2	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO5	PO9	PO10	PSO1
	2	2	1	1	1	1
Justification for correlation	Apply Engineering fundamentals knowledge to find computational solutions to various problems.	To review concepts of GPRS architecture	Apply appropriate techniques, resources to complex engineering activities with an understanding the architecture.	To work as an individual	To Communicate effectively on complex engineering activities	Will be able to solve complex computational task

- Images / Screenshot of the practice:



Fig 1: Students individually prepared and writing the given question



Mobile Computing
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Name: Saravana Jothi G

GPRS

- * GPRS stands for General Packet Radio Services.
- * GPRS when integrated with GSM improves internet access.
- * GPRS transfers packet from GSM mobile network to external data network.
- * 'Packet switched' network is used in GPRS.
- * In GPRS, billing system is based on amount of data transferred.
- * GPRS end to end packet switched services are categorized into two services.
 - Point to point service
 - Point to multipoint service

Upgrades of GSM network elements for GPRS

- 1) Mobile Station - New mobile station is required to access GPRS.
- 2) BTS - Base Transceiver system requires software upgrade.
- 3) BSC - Base Station Controller requires new hardware unit called 'Packet Control Unit'.
- 4) GPRS support nodes - Network elements like SGSN (Serving GPRS support node) and GGSN (Gateway GPRS support node) are required.

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① GPRS:

- * When GPRS is integrated with GSM, it significantly improves & simplifies internet access.
- * It transfers data packets from GSM mobile stations to external data networks.
- * Packets are routed from GSM mobile stations to packet switch networks which makes it easy to connect with the internet.
- * It offers end-to-end packet switched data transfer services which are categorized into,
 - Point-to-Point service (PTP)
 - Multipoint → Point to Multipoint (PTM) service

PTP:

It is between two users and can either be connectionless or connection oriented.

PTM:

It is a data transfer service from one user to multiple users.

- Multicast PTM
- Group call PTM

- * GPRS is packet based mobile data service on the data service.
- * It is a non voice, high-speed and packet switching technology.

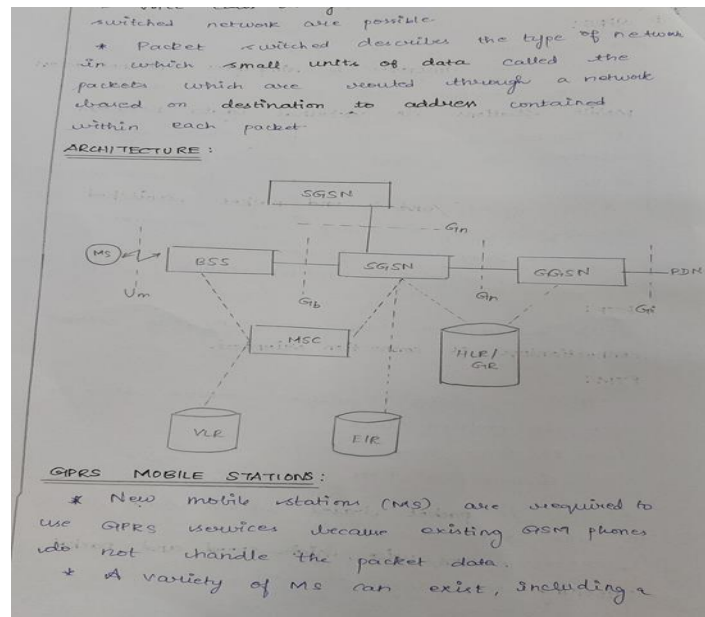


Fig 2: Sample One Minute Paper of Saravanajothi G and Venesha L.



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students informed the instructor that the activity motivates them to pay attention in class and ask questions. Also they can able to recall the key terms in each component.

- ❖ ***Benefit of the practice:***

- Students were actively participated in this activity.
- From this activity, the students can get more clarity in the particular topic.
- Students can able to explain the concepts without any confusion.
- Able to attend this topic in Internal Assessment Test II.

- ❖ ***Challenges faced in implementation:***

- Some students are not understanding the concepts and they were not completed the task.
- Students were asked the extra time to complete the questions and the were not completing within 10 Minutes.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/IG_CS8494-One_Minute_Paper.pdf <https://oncourseworkshop.com/self-awareness/one-minute-paper/> <https://www.eur.nl/en/teacheur/one-minute-paper>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/2021-2022/Unit-3%20One%20Minute.pdf>

Signature of Faculty Member

HOD-CSE